Tutorial 2 Solutions

Exercise 1

- **a.** There are **3 records** and **6 fields** present in this table.
- **b.** Field classifications:
  - `product_code` : **Text** (or Alpha-numeric String)
  - `product_name` : **Text** (or String)
  - `supplier_cost` : **Number** (or Decimal/Float/Currency)
  - `clearance_item` : **Boolean** (logical values: True/False)
  - `warehouse_zone` : **Text** (or String)
  - `reorder_date` : **Date**

Exercise 2

- **a. Field Identification Table:**
  1. `nurse_id` , `overtime_hours` , `hourly_overtime_rate` , `pay_date`
  2. `supplier_id` , `item_category` , `delivery_date`
  3. `vault_id` , `item_unit_cost` , `quantity_on_hand`
  4. `medication_id` , `batch_expiration_date` , (`medication_name`)
  5. `tool_category` , `maintenance_cost` , (`service_date`)

Exercise 3

N	Example of Activity	Intranet	Extranet	Internet
1	An accountant updating an internal branch-specific expense ledger.	✓		
2	Giving trusted wholesale dealerships a portal to check available vehicle stock.		✓	
3	A prospective client submitting an inquiry form via the public website.			✓
4	Uploading a team performance review template to the HR shared drive.	✓		
5	A vendor uploading compliance certificates directly to a company's purchasing hub.		✓	

Exercise 4:

Task Description	SDLC Phase
1. Running isolated stress tests on newly written database sub-routines.	**Tests**
2. Meeting with bank compliance officers to document regulatory security boundaries.	**Preliminary Analysis / Requirements gathering**
3. Finalizing structural wireframes, data dictionaries, and architectural maps.	**System Design**
4. Migrating historical user accounts and hosting technical training workshops for staff.	**Deployment**
5. Writing modular, well-documented backend microservices in Python or C.	**Programming**

Exercise 5:

Description	Term / Methodology
1. An iterative approach characterized by deploying small, frequent functional updates within strictly bound, predictable timeboxes.	**Agile**
2. A structural shift or strategy course-correction executed by software product owners after analyzing real-world metrics from a lean release.	**Pivot**
3. A traditional predictive development framework that is highly structured and best optimized for highly stable, deterministic enterprise systems.	**SDLC (Waterfall)**
4. A delivery approach where the absolute end-state system specifications cannot be mapped out or finalized during the initial project phases.	**Lean Methodology**
5. A fast-tracked development methodology relying heavily on interactive visual prototyping cycles directly involving target end-users.	**RAD (Rapid Application Development)**

Exercise 6:

1. **Integrated Development Environment (IDE)** → **C.** A unified engineering workspace combining text manipulation tools, semantic syntax validation, compilers, and internal execution tracers.
2. **2. Integration Testing** → **D.** A structured process of testing multiple combined software modules together to ensure they interact correctly and pass data seamlessly.
3. **CASE Tools** → **A.** Specialized suites engineered to automate code generation, model database designs visually, and streamline system analysis without low-level manual script-writing.
4. **Minimum Viable Product (MVP)** → **B.** A foundational, functional application build deployed rapidly to real-world target environments to collect maximum validated feedback with minimal effort.